

Boolean Antichains in Finite Lattices

Complete Classification, Product Structure, and Matroid Applications

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Abstract

Garber, Goltermann, Horiatakis, König, and Gottesman asked for constructions and counts of Boolean antichains in geometric lattices and proposed a spanning-tree correspondence for graphic matroids. We solve these problems in greater generality by classifying Boolean antichains of every size in every finite lattice. For an ordered family, intersecting all but one member produces candidate Boolean atoms. The family is Boolean precisely when those atoms rise above the common meet and all 2^k reconstruction height gaps vanish. This gives an exact finite formula for the complete enumerator. Its binomial transform is multiplicative under lattice products, equivalently giving an explicit positive convolution.

For a rank- r matroid, maximum Boolean antichains in the flat lattice are naturally in bijection with bases of the simplification. For graphs this gives spanning forests and specializes to the proposed spanning-tree bijection in the connected simple case. Retaining parallel classes recovers the multivariate basis polynomial, while flat intervals give minor-local and rank-tight versions. The framework also yields explicit all-size formulas for uniform matroids, complete classifications for finite distributive and subspace lattices, a universal Möbius formula for Boolean pairs, and complete distributions for finite projective planes.

1 Introduction

Let L be a finite lattice. A Boolean antichain is an antichain C for which the map

$$S \mapsto \bigwedge S$$

is a lattice anti-isomorphism from the power set of C to the sublattice generated by C and the greatest element of L . This definition was introduced for representation-theoretic purposes and was recently studied enumeratively by Garber, Goltermann, Horiatakis, König, and Gottesman [1]. Among other results, they proved that Boolean antichains of size $n - 1$ in the partition lattice are counted by the labelled trees on $[n]$. Since the partition lattice is the lattice of flats of the graphic matroid of K_n , they expected the construction to extend to an arbitrary graph and its spanning trees. They also asked more generally for constructions and counts in geometric lattices.

The main result is lattice-wide. Given an ordered tuple $\mathbf{H} = (H_1, \dots, H_k)$, let $X_{\mathbf{H}}$ be its common meet and let $A_i(\mathbf{H})$ be the meet of all entries except H_i . For $S \subseteq [k]$, compare the join of the A_i with $i \in S$ to the meet of the complementary H_j . The former always lies below the latter. Their height difference is the reconstruction gap $\gamma_{\mathbf{H}}(S)$. Theorem 3.1 proves that $\mathbf{H}$ orders a Boolean antichain precisely when every A_i rises above $X_{\mathbf{H}}$ and every reconstruction gap vanishes.

Consequently,

$$|\mathcal{C}_k(L)| = \frac{1}{k!} \sum_{\mathbf{H} \in L^k} \prod_i \mathbf{1}_{\{\text{ht}_L A_i(\mathbf{H}) > \text{ht}_L X_{\mathbf{H}}\}} \prod_{S \subseteq [k]} \mathbf{1}_{\{\gamma_{\mathbf{H}}(S)=0\}}.$$

Thus the complete enumerator is determined for every finite lattice, without a distributivity, modularity, or grading hypothesis.

The second structural result describes products. If $c_k(L) = |\mathcal{C}_k(L)|$ and

$$\eta_k(L) = \sum_{a=0}^k \binom{k}{a} a! c_a(L),$$

then Theorem 3.2 gives

$$\eta_k(L \times K) = \eta_k(L) \eta_k(K).$$

Equivalently, the coefficients satisfy an explicit positive convolution. This product law recovers the Stirling recurrence for Boolean lattices and computes every layer of a matroid direct sum from its summands.

The motivating geometric-lattice problem is the principal application of this framework. Every geometric lattice is the lattice of flats of a simple matroid [3]. The subsets of a matroid basis are known to span a Boolean sublattice [4]; the new point is the inverse classification and its enumerative refinements. If M has rank r , the atoms of any size- r Boolean span are rank-one flats forming a basis of $\text{si}(M)$, and the construction is reversible. Hence

$$\{\text{Boolean antichains of size } r \text{ in } \mathcal{L}(M)\} \longleftrightarrow \{\text{bases of } \text{si}(M)\}.$$

For a graph, this is the spanning-forest bijection; the connected simple case is exactly the spanning-tree extension proposed by Garber et al.

The flat lattice remembers each parallel class as a single atom. Retaining the class recovers the full multivariate basis polynomial:

$$\sum_{B \in \mathcal{B}(M)} \prod_{e \in B} x_e = \sum_C \prod_{A \in \text{At}(\langle\langle C \rangle\rangle)} \left(\sum_{e \in A \setminus \text{cl}(\emptyset)} x_e \right).$$

The same statement holds in every flat interval through the minor $(M|Y)/X$, yielding the rank-tight classification by contraction bases. The global height-gap theorem supplies all remaining layers, whose face ranks are strict integral polymatroid ranks.

Several families admit more compact evaluations. Boolean pairs have a universal Möbius formula; finite distributive lattices are described by disjoint order filters; finite subspace lattices are described by quotient direct-sum decompositions; and uniform matroids and finite projective planes have explicit all-size formulas. Section 2 introduces the motivating flat-lattice notation. Section 3 proves the global height-gap and product theorems. The remaining sections develop the universal specializations and the matroid applications.

2 Matroid motivation and flat-lattice notation

Let M be a finite matroid on ground set E , with rank function rk_M and closure operator cl_M . Its lattice of flats is denoted by $\mathcal{L}(M)$. The meet is intersection and the join is

$$F \vee G = \text{cl}_M(F \cup G).$$

The least flat is $L_0 = \text{cl}_M(\emptyset)$, the set of loops, and the greatest flat is E . A maximal chain adds one rank at each cover, so the height of $\mathcal{L}(M)$ is $r = \text{rk}(M)$.

The simplification $\text{si}(M)$ deletes the loops and replaces each nonloop parallel class by a single element. We identify its ground set with the rank-one flats of M : a rank-one flat A represents the parallel class $A^\circ = A \setminus L_0$. A basis of $\text{si}(M)$ is therefore a set $\{A_1, \dots, A_r\}$ of rank-one flats whose join is E .

For a Boolean antichain $C = \{H_1, \dots, H_k\}$, write

$$\langle C \rangle = \left\{ \bigwedge_{i \in S} H_i : S \subseteq [k] \right\}, \quad \bigwedge \emptyset = E.$$

Its bottom is $X_C = \bigcap_i H_i$. The atoms of the Boolean lattice $\langle C \rangle$ are

$$A_i(C) = \bigcap_{j \neq i} H_j. \tag{1}$$

Lemma 2.1 (Full-height rigidity). *Let L be a ranked lattice of rank r , and let C be a Boolean antichain of size r . Then the bottom of $\langle C \rangle$ is the bottom of L , and every maximal chain of $\langle C \rangle$ is a maximal chain of L . In particular, the atoms and coatoms of $\langle C \rangle$ have ambient ranks 1 and $r - 1$, respectively.*

Proof. A maximal chain in $\langle C \rangle \cong B_r$ has r cover steps. Viewed in L , each strict step raises rank by at least one. Its top has rank r , so its bottom has rank zero and every step raises rank by one. A ranked lattice has a unique element of rank zero, and the remaining assertions follow. $\square$

We will also use the following familiar basis fact in a form that records meets as well as joins.

Lemma 2.2 (Boolean span of a simplified basis). *Let $\mathcal{A} = \{A_1, \dots, A_r\}$ be a basis of $\text{si}(M)$. For $S \subseteq [r]$, put $F_S = \bigvee_{i \in S} A_i$, with $F_\emptyset = L_0$. Then*

$$F_S \vee F_T = F_{S \cup T}, \quad F_S \cap F_T = F_{S \cap T}.$$

Consequently $S \mapsto F_S$ embeds B_r as a sublattice of $\mathcal{L}(M)$.

Proof. The join identity is immediate. Choose one representative from each A_i° . These representatives form a basis of M , so $\text{rk}(F_S) = |S|$. Submodularity gives

$$\text{rk}(F_S \cap F_T) \leq |S| + |T| - |S \cup T| = |S \cap T|.$$

The reverse inequality follows from $F_{S \cap T} \subseteq F_S \cap F_T$. The two flats have the same rank and one contains the other, hence they are equal. $\square$

3 Global height-gap classification

Let L now be an arbitrary finite lattice, and let $\text{ht}_L(Z)$ be the maximum length of a chain from its bottom to Z . Strict order implies strict increase of ht_L . The preceding definition of $A_i(C)$ makes sense for an arbitrary ordered tuple $\mathbf{H} = (H_1, \dots, H_k) \in L^k$, whether or not it is Boolean. Put

$$X_{\mathbf{H}} = \bigwedge_{i=1}^k H_i, \quad A_i(\mathbf{H}) = \bigwedge_{j \neq i} H_j.$$

For $S \subseteq [k]$, define two candidate faces

$$U_{\mathbf{H}}(S) = X_{\mathbf{H}} \vee \bigvee_{i \in S} A_i(\mathbf{H}),$$

$$V_{\mathbf{H}}(S) = \bigwedge_{j \notin S} H_j,$$

where the empty relative join is $X_{\mathbf{H}}$ and the empty meet is $\hat{1}$. Every $A_i(\mathbf{H})$ lies below every H_j with $i \neq j$, and hence

$$U_{\mathbf{H}}(S) \leq V_{\mathbf{H}}(S).$$

The *reconstruction height gap* of S is the nonnegative integer

$$\gamma_{\mathbf{H}}(S) = \text{ht}_L V_{\mathbf{H}}(S) - \text{ht}_L U_{\mathbf{H}}(S). \quad (2)$$

Theorem 3.1 (Global height-gap criterion). *Let L be a finite lattice and let $\mathbf{H} = (H_1, \dots, H_k) \in L^k$. The following statements are equivalent:*

1. $\mathbf{H}$ is an ordering of a size- k Boolean antichain;
2. the conditions

$$\text{ht}_L A_i(\mathbf{H}) > \text{ht}_L X_{\mathbf{H}} \quad \text{for every } i \in [k], \quad (3)$$

$$\gamma_{\mathbf{H}}(S) = 0 \quad \text{for every } S \subseteq [k]. \quad (4)$$

hold.

Consequently, writing $\mathbf{1}_{\{P\}}$ for the indicator of a proposition P , the exact count is

$$|\mathcal{C}_k(L)| = \frac{1}{k!} \sum_{\mathbf{H} \in L^k} \prod_{i=1}^k \mathbf{1}_{\{\text{ht}_L A_i(\mathbf{H}) > \text{ht}_L X_{\mathbf{H}}\}} \prod_{S \subseteq [k]} \mathbf{1}_{\{\gamma_{\mathbf{H}}(S)=0\}}. \quad (5)$$

This formula includes $|\mathcal{C}_0(L)| = 1$ and is zero for $k > \text{ht}_L(\hat{1})$.

Proof. Suppose first that $\mathbf{H}$ orders a Boolean antichain. In its Boolean span, $X_{\mathbf{H}}$ is the bottom and the $A_i(\mathbf{H})$ are the atoms, so Equation (3) holds. The face indexed by S is both the join of its atoms and the meet of the complementary coatoms. Thus $U_{\mathbf{H}}(S) = V_{\mathbf{H}}(S)$ and every reconstruction gap vanishes.

Conversely, assume Equations (3) and (4). Since $U_{\mathbf{H}}(S) \leq V_{\mathbf{H}}(S)$, equality of their heights implies equality of the two elements. Write their common value as F_S . Then

$$F_S \wedge F_T = V_{\mathbf{H}}(S) \wedge V_{\mathbf{H}}(T) = V_{\mathbf{H}}(S \cap T) = F_{S \cap T},$$

$$F_S \vee F_T = U_{\mathbf{H}}(S) \vee U_{\mathbf{H}}(T) = U_{\mathbf{H}}(S \cup T) = F_{S \cup T}.$$

It remains to prove that the 2^k faces are distinct. If $S \not\subseteq T$, choose $i \in S \setminus T$. Then $A_i(\mathbf{H}) \leq F_S$ and $F_T \leq H_i$. Equality $F_S = F_T$ would imply $A_i(\mathbf{H}) \leq H_i$, whereas

$$A_i(\mathbf{H}) \wedge H_i = X_{\mathbf{H}}.$$

This contradicts Equation (3). Hence $S \mapsto F_S$ is a lattice embedding of B_k , and $F_{[k] \setminus \{i\}} = H_i$. Its image is exactly the sublattice generated by the H_i and $\hat{1}$, so $\mathbf{H}$ orders a Boolean antichain.

The conditions also force every H_i to be proper and the entries to be distinct: a top or repeated entry would make some A_i equal to $X_{\mathbf{H}}$. Thus the symmetric group acts freely on the admissible tuples, and every unordered Boolean antichain has exactly $k!$ orderings. This proves Equation (5). The height bound follows from the strict chain $F_{\emptyset} < F_{\{1\}} < \dots < F_{[k]}$. $\square$

Only 2^k height gaps occur in Equation (5); the gaps for $S = \emptyset$ and $|S| = 1$ vanish identically. Thus the criterion gives an exact finite algorithm using $O(2^k)$ lattice operations per ordered tuple. This replaces the pairwise test on all 2^k generated faces.

The full enumerator also has an exact product law. Write $c_k(L) = |\mathcal{C}_k(L)|$, $e_k(L) = k!c_k(L)$, and

$$\eta_k(L) = \sum_{a=0}^k \binom{k}{a} e_a(L). \quad (6)$$

Theorem 3.2 (Lattice-product convolution). *For finite lattices L and K ,*

$$\eta_k(L \times K) = \eta_k(L)\eta_k(K). \quad (7)$$

Equivalently,

$$c_k(L \times K) = \sum_{\substack{0 \leq a, b \leq k \\ a+b \geq k}} \frac{a!b!}{(k-a)!(k-b)!(a+b-k)!} c_a(L)c_b(K). \quad (8)$$

Proof. The quantity $e_a(L)$ counts labelled lattice embeddings $B_a \hookrightarrow L$ that preserve the top. A top-preserving lattice homomorphism $\phi : B_k \rightarrow L$ may collapse some Boolean atoms to the image of the bottom. If $A \subseteq [k]$ is the set of uncollapsed atoms, the restriction to those atoms is a labelled embedding of $B_{|A|}$. Indeed, if two subsets of A had the same image, an atom in their symmetric difference would lie below that common image while meeting it in the image of the bottom, forcing that atom to be collapsed. The collapsed atoms do not change joins, so this embedding still preserves the top of L . Conversely, a top-preserving embedding on A extends uniquely by sending a subset of $[k]$ to the image of its intersection with A . Hence $\eta_k(L)$ counts all top-preserving homomorphisms $B_k \rightarrow L$.

A homomorphism to $L \times K$ is a pair of coordinate homomorphisms, proving Equation (7). The product map is injective precisely when the two active sets $A, B \subseteq [k]$ cover $[k]$. For $|A| = a$, $|B| = b$, the number of such pairs is

$$\frac{k!}{(k-a)!(k-b)!(a+b-k)!}.$$

Multiplying by $e_a(L)e_b(K)$, summing, and dividing by $k!$ gives Equation (8). $\square$

For matroids, $\mathcal{L}(M_1 \oplus M_2) = \mathcal{L}(M_1) \times \mathcal{L}(M_2)$, so Equation (8) computes every Boolean-antichain layer of a direct sum from those of its summands. Taking $K = B_1$ gives

$$c_k(L \times B_1) = (k+1)c_k(L) + c_{k-1}(L).$$

Iteration from the one-element lattice recovers $c_k(B_n) = S(n+1, k+1)$ together with the Stirling recurrence.

For $L = \mathcal{L}(M)$, lattice height is matroid rank. Thus Theorem 3.1 specializes to a rank-gap certificate in every finite geometric lattice.

The ambient ranks of the reconstructed faces retain additional structure.

Corollary 3.3 (Rank profiles). *For $C \in \mathcal{C}_k(M)$, put*

$$\rho_C(S) = \text{rk}_M(F_S) - \text{rk}_M(X_C).$$

Then ρ_C is a normalized, strictly increasing integral polymatroid rank function on $[k]$. Its rank-tight members are characterized by $\rho_C(S) = |S|$ for every $S \subseteq [k]$.

Proof. Normalization and strict increase follow from $F_\emptyset = X_C$ and the strict Boolean embedding. Matroid submodularity and $F_S \cap F_T = F_{S \cap T}$, $F_S \vee F_T = F_{S \cup T}$ give

$$\rho_C(S) + \rho_C(T) \geq \rho_C(S \cap T) + \rho_C(S \cup T).$$

If C is rank-tight, every one of the k strict steps in a maximal Boolean chain raises ambient rank by one, giving $\rho_C(S) = |S|$. The converse is immediate at $S = [k]$. $\square$

The criterion also turns the full uniform-matroid problem into a closed block count.

Theorem 3.4 (All sizes in a uniform matroid). *Let $1 \leq r \leq n$. Then $|\mathcal{C}_0(U_{r,n})| = 1$ and*

$$|\mathcal{C}_1(U_{r,n})| = \sum_{b=0}^{r-1} \binom{n}{b}.$$

For $2 \leq k \leq r$, write $p = p_1 + \dots + p_k$. Then

$$|\mathcal{C}_k(U_{r,n})| = \frac{1}{k!} \sum_{b=0}^{r-1} \binom{n}{b} \sum_{\substack{p_1, \dots, p_k \geq 1, \ p \leq n-b \\ b+p \geq r, \ b+p-p_i < r \ (i \in [k])}} \frac{(n-b)!}{(n-b-p)! \prod_{i=1}^k p_i!}. \quad (9)$$

In particular, the size distributions of $U_{3,4}$ and $U_{4,5}$ are respectively

$$(1, 11, 27, 4), \quad (1, 26, 150, 50, 5).$$

Proof. For $k = 1$, every proper flat is a Boolean antichain, and the proper flats of $U_{r,n}$ are the subsets of size less than r .

Let $k \geq 2$ and let X and A_i be the bottom and atoms of a Boolean span. All are proper uniform-matroid flats. Since $A_i \cap A_j = X$ for $i \neq j$, the sets $P_i = A_i \setminus X$ are pairwise disjoint and nonempty. Put $b = |X|$ and $p_i = |P_i|$. Every proper Boolean face is the ordinary union

$$X \cup \bigcup_{i \in S} P_i.$$

The full-join and proper-coatom conditions are respectively $b + p \geq r$ and $b + p - p_i < r$ for every i ; these inequalities then make every proper face a flat. Conversely, any such bottom and blocks produce all the faces of a Boolean sublattice, so this construction is bijective.

There are $\binom{n}{b}$ choices for X . For fixed ordered block sizes, the number of ordered disjoint blocks is

$$\frac{(n-b)!}{(n-b-p)! \prod_i p_i!}.$$

Division by the free permutation action on the k nonempty blocks gives Equation (9). $\square$

4 Two-element Boolean antichains

The first off-diagonal layer has a closed form in every finite lattice, not only in geometric lattices. For a finite lattice L , put

$$\ell(X) = |[\hat{0}, X]|,$$

and let μ_L be its Möbius function.

Theorem 4.1 (Möbius formula for pairs). *Two elements F, G form a Boolean antichain if and only if both are proper and $F \vee G = \hat{1}$. Consequently,*

$$|\mathcal{C}_2(L)| = \frac{1}{2} \left(\sum_{X \leq \hat{1}} \mu_L(X, \hat{1}) \ell(X)^2 - 2|L| + 1 \right). \quad (10)$$

Proof. If $F, G < \hat{1}$ and $F \vee G = \hat{1}$, then neither can contain the other. The four elements

$$\hat{1}, \quad F, \quad G, \quad F \wedge G$$

are distinct and form the Boolean span prescribed by the meet map. The converse follows because the join of the two Boolean coatoms is the top.

For $Y \in L$, let $g(Y)$ be the number of ordered pairs (F, G) with $F \vee G = Y$. The number of ordered pairs whose join is at most Y is $\ell(Y)^2$, so

$$\ell(Y)^2 = \sum_{X \leq Y} g(X).$$

Möbius inversion gives

$$g(\hat{1}) = \sum_{X \leq \hat{1}} \mu_L(X, \hat{1}) \ell(X)^2.$$

Among these ordered pairs, exactly $2|L| - 1$ have at least one coordinate equal to $\hat{1}$. Every remaining pair consists of incomparable proper elements and is counted twice as an unordered Boolean antichain. This proves Equation (10). $\square$

For $L = B_n$, one has $\ell(X) = 2^{\text{rk}(X)}$ and $\mu(X, \hat{1}) = (-1)^{n-\text{rk}(X)}$. The sum in Equation (10) is therefore 3^n , and

$$|\mathcal{C}_2(B_n)| = \frac{3^n - 2^{n+1} + 1}{2} = S(n+1, 3),$$

recovering the size-two instance of the source formula.

5 All sizes in distributive lattices

Let P be a finite poset and let $J(P)$ be its distributive lattice of order ideals. Denote by $\mathcal{F}^+(P)$ the set of nonempty order filters of P , and let Γ_P be their intersection graph: two filters are adjacent when they have nonempty intersection. Write

$$I(\Gamma_P; t) = \sum_{k \geq 0} i_k(\Gamma_P) t^k$$

for its independence polynomial.

Theorem 5.1 (Distributive-lattice classification). *Complementation gives a size-preserving bijection*

$$\mathcal{C}_k(J(P)) \longleftrightarrow \{ \{F_1, \dots, F_k\} \subseteq \mathcal{F}^+(P) : F_i \cap F_j = \emptyset \text{ for } i \neq j \}.$$

Consequently,

$$\sum_{k \geq 0} |\mathcal{C}_k(J(P))| t^k = I(\Gamma_P; t). \quad (11)$$

Proof. A Boolean span has pairwise top joins. Conversely, suppose that a family of proper elements in a distributive lattice has pairwise top joins. Finite distributivity gives, for subsets S, T of the family,

$$\left(\bigwedge S\right) \vee \left(\bigwedge T\right) = \bigwedge_{a \in S, b \in T} (a \vee b) = \bigwedge (S \cap T),$$

because the factors with $a \neq b$ are the top and the remaining factors are the common elements. This is the defining meet-map identity.

Complementation sends the proper ideals bijectively to the nonempty order filters. Moreover,

$$I_i \cup I_j = P \iff (P \setminus I_i) \cap (P \setminus I_j) = \emptyset.$$

Thus complementation is the displayed bijection. Pairwise disjoint filters are independent sets of Γ_P , proving Equation (11). $\square$

If P is an n -element antichain, then its nonempty filters are the nonempty subsets of P . An independent set in Γ_P is an unordered family of disjoint nonempty subsets; adjoining the unused elements as one distinguished block gives

$$|\mathcal{C}_k(B_n)| = S(n+1, k+1).$$

If P is an n -element chain, all nonempty filters intersect, so the only nonzero counts are $|\mathcal{C}_0| = 1$ and $|\mathcal{C}_1| = n$.

6 Maximum Boolean antichains

Let $\mathcal{C}_r(M)$ denote the set of Boolean antichains of size r in $\mathcal{L}(M)$, and let $\mathcal{B}(\text{si}(M))$ denote the set of bases of the simplification.

Theorem 6.1 (Maximum-antichain bijection). *For every finite rank- r matroid M , the maps*

$$\begin{aligned} \Phi : \mathcal{C}_r(M) &\longrightarrow \mathcal{B}(\text{si}(M)), & C &\longmapsto \{A_1(C), \dots, A_r(C)\}, \\ \Psi : \mathcal{B}(\text{si}(M)) &\longrightarrow \mathcal{C}_r(M), & \{A_1, \dots, A_r\} &\longmapsto \left\{ \bigvee_{j \neq i} A_j : i \in [r] \right\} \end{aligned}$$

are well defined and inverse to one another. Hence

$$|\mathcal{C}_r(M)| = |\mathcal{B}(\text{si}(M))| = T_{\text{si}(M)}(1, 1). \quad (12)$$

Proof. Let $C = \{H_1, \dots, H_r\}$ lie in $\mathcal{C}_r(M)$. By Lemma 2.1, the flats $A_i(C)$ in Equation (1) have rank one. Their join is the top of the Boolean span, namely E . Thus they are r rank-one flats of total rank r , so they form a basis of $\text{si}(M)$. This proves that Φ is well defined.

Conversely, let $\mathcal{A} = \{A_1, \dots, A_r\}$ be a basis of $\text{si}(M)$. Lemma 2.2 shows that its joins form a Boolean sublattice. The displayed family defining $\Psi(\mathcal{A})$ is precisely the set of coatoms of this sublattice. The power-set meet map is therefore an anti-isomorphism, so the family is a Boolean antichain.

The atoms of the Boolean span of $\Psi(\mathcal{A})$ are exactly the A_i , and the coatoms of the Boolean span of C are exactly the H_i . Therefore $\Phi\Psi$ and $\Psi\Phi$ are the identity maps. The last equality in Equation (12) is the standard basis evaluation of the Tutte polynomial. $\square$

The theorem resolves the graphic expectation of Garber et al. at matroid scope. Its exact form records the necessary conventions: a nonsimple matroid is counted after simplification, and the relevant size is the matroid rank.

7 The multivariate basis polynomial

The bijection loses no information about the bases of the original matroid once its rank-one flats are retained. Define the multivariate basis polynomial

$$\mathbf{B}_M(\mathbf{x}) = \sum_{B \in \mathcal{B}(M)} \mathbf{x}^B, \quad \mathbf{x}^B = \prod_{e \in B} x_e.$$

Theorem 7.1 (Parallel-class decomposition). *For every finite matroid M ,*

$$\boxed{\mathbf{B}_M(\mathbf{x}) = \sum_{C \in \mathcal{C}_r(M)} \prod_{A \in \text{At}(\langle C \rangle)} \left(\sum_{e \in A \setminus L_0} x_e \right)}. \quad (13)$$

In particular, if $b(M)$ is the number of bases of M , then

$$b(M) = \sum_{C \in \mathcal{C}_r(M)} \prod_{A \in \text{At}(\langle C \rangle)} |A \setminus L_0|. \quad (14)$$

Proof. Every basis B of M meets exactly r distinct nonloop parallel classes. Their rank-one flats form a basis of $\text{si}(M)$ and hence, by Theorem 6.1, determine a unique $C \in \mathcal{C}_r(M)$. Conversely, for the atoms of the Boolean span of a fixed C , every choice of one element from each $A \setminus L_0$ is a basis of M . Thus the bases are partitioned by C , and expanding the corresponding product of linear forms gives Equation (13). Setting every variable equal to one gives Equation (14). $\square$

For a simple matroid every factor in Equation (14) is one, reducing it to the bijective count. The polynomial form is the correct statement for multigraphs and other parallel extensions.

8 Intervals, minors, and rank-tight antichains

Let $X \leq Y$ be flats of M . The interval $[X, Y]$ is canonically the lattice of flats of

$$N = (M|Y)/X.$$

Indeed, the correspondence sends a flat F with $X \subseteq F \subseteq Y$ to $F \setminus X$; the contraction rank identity

$$\text{rk}_N(S) = \text{rk}_M(S \cup X) - \text{rk}_M(X)$$

shows directly that it preserves closed sets and inclusion. A Boolean antichain in this interval is defined relative to its top Y . Write $d = \text{rk}_M(Y) - \text{rk}_M(X)$.

Theorem 8.1 (Interval form). *The Boolean antichains of size d in $[X, Y]$ are naturally in bijection with the bases of $\text{si}((M|Y)/X)$. Moreover,*

$$\mathbf{B}_{(M|Y)/X}(\mathbf{x}) = \sum_C \prod_{A \in \text{At}(\langle C \rangle)} \left(\sum_{e \in A \setminus X} x_e \right), \quad (15)$$

where C runs over the size- d Boolean antichains in $[X, Y]$.

Proof. Apply Theorems 6.1 and 7.1 to the minor N under the flat-lattice isomorphism $\mathcal{L}(N) \cong [X, Y]$. The rank-one flat of N corresponding to an interval atom A has parallel class $A \setminus X$, which gives the displayed factors. $\square$

Definition 8.2. A Boolean antichain C of size k in $\mathcal{L}(M)$ is *rank-tight* if

$$\text{rk}_M(X_C) = r - k.$$

Equivalently, its Boolean span has the full height of the interval $[X_C, E]$.

Corollary 8.3 (Rank-tight enumeration). *The rank-tight Boolean antichains of size k are partitioned by their bottom flat, and*

$$\#\{C : |C| = k, C \text{ rank-tight}\} = \sum_{\substack{X \in \mathcal{L}(M) \\ \text{rk}_M(X) = r - k}} b(\text{si}(M/X)). \quad (16)$$

For each fixed X , the bijection is the interval bijection of Theorem 8.1 on $[X, E]$.

Proof. If C is rank-tight with bottom X , then it is a maximum Boolean antichain in the rank- k interval $[X, E]$. Conversely, every maximum Boolean antichain in that interval is rank-tight in $\mathcal{L}(M)$. Apply Theorem 8.1 and sum over the possible bottom flats. $\square$

9 Closed rank-tight enumerations

Corollary 8.3 becomes completely explicit for the Boolean and partition lattices treated in the source paper.

Corollary 9.1 (Boolean lattices). *The number of rank-tight Boolean antichains of size k in the Boolean lattice B_n is*

$$\binom{n}{k}. \quad (17)$$

Since the total number of size- k Boolean antichains is $S(n+1, k+1)$ [1], the number of non-rank-tight ones is

$$S(n+1, k+1) - \binom{n}{k}. \quad (18)$$

Proof. The lattice B_n is the flat lattice of the free matroid on n elements. A corank- k flat is the complement of a k -subset, and the contraction by that flat is the free matroid on those k elements. Its simplification has one basis. Equation (17) follows from Corollary 8.3; subtract it from the total count for the last assertion. $\square$

Corollary 9.2 (Partition lattices). *For $1 \leq k \leq n-1$, the number of rank-tight Boolean antichains of size k in the partition lattice Π_n is*

$$S(n, k+1)(k+1)^{k-1}. \quad (19)$$

At $k = n-1$, this is Cayley's value n^{n-2} for maximum Boolean antichains.

Proof. Identify Π_n with $\mathcal{L}(M(K_n))$. A corank- k flat is a set partition of $[n]$ into $k+1$ blocks, of which there are $S(n, k+1)$. Contracting the flat collapses each block to a vertex. Between every two resulting vertices there is a nonempty parallel class, so the simplification of the contraction is $M(K_{k+1})$. It has $(k+1)^{k-1}$ bases by Cayley's formula. Apply Corollary 8.3. $\square$

The same calculation gives all rank-tight layers of a uniform matroid.

Corollary 9.3 (Uniform matroids). *For $2 \leq r \leq n$, the number $R_{r,n}(k)$ of rank-tight size- k Boolean antichains in $\mathcal{L}(U_{r,n})$ is*

$$R_{r,n}(k) = \begin{cases} 1, & k = 0, \\ \binom{n}{r-1}, & k = 1, \\ \binom{n}{r} \binom{r}{k}, & 2 \leq k \leq r. \end{cases}$$

Proof. A corank- k flat has size $r - k$. For $k = 1$, the rank-one contraction simplifies to a one-element matroid. For $k \geq 2$, the contraction is $U_{k,n-r+k}$, which is simple and has $\binom{n-r+k}{k}$ bases. Thus Corollary 8.3 gives

$$\binom{n}{r-k} \binom{n-r+k}{k} = \binom{n}{r} \binom{r}{k}.$$

□

The same Möbius formula gives a compact expression for the entire size-two layer of the partition lattice, including the non-rank-tight part.

Corollary 9.4 (All Boolean pairs in Π_n). *Let B_m be the m th Bell number and put*

$$A(z) = \sum_{m \geq 1} B_m^2 \frac{z^m}{m!}.$$

Then, for $n \geq 1$,

$$|\mathcal{C}_2(\Pi_n)| = \frac{1}{2} (n! [z^n] \log(1 + A(z)) - 2B_n + 1). \quad (20)$$

The initial values for $n = 1, 2, \dots$ are

$$0, 0, 3, 45, 620, 9750, 183680, \dots$$

Proof. For a set partition π , the lower interval is the product of the partition lattices on its blocks, so

$$\ell(\pi) = \prod_{D \in \pi} B_{|D|}.$$

If $b(\pi)$ is the number of blocks, then $\mu(\pi, \hat{1}) = (-1)^{b(\pi)-1} (b(\pi) - 1)!$. The exponential formula now gives

$$\sum_{\pi \in \Pi_n} \mu(\pi, \hat{1}) \ell(\pi)^2 = n! [z^n] \sum_{b \geq 1} \frac{(-1)^{b-1}}{b} A(z)^b = n! [z^n] \log(1 + A(z)).$$

Substitute this and $|\Pi_n| = B_n$ into Equation (10). □

10 All sizes in subspace lattices

Let $V = \mathbb{F}_q^n$, where q is a prime power, and let $\mathcal{S}(V)$ be its lattice of linear subspaces. Write $\begin{bmatrix} n \\ d \end{bmatrix}_q$ for the Gaussian binomial coefficient and

$$g_d(q) = |\mathrm{GL}_d(q)| = \prod_{i=0}^{d-1} (q^d - q^i), \quad g_0(q) = 1.$$

For $1 \leq k \leq d$, put

$$D_{d,k}(q) = \frac{1}{k!} \sum_{\substack{d_1 + \dots + d_k = d \\ d_i \geq 1}} \frac{g_d(q)}{\prod_{i=1}^k g_{d_i}(q)}. \quad (21)$$

Theorem 10.1 (Subspace-lattice enumeration). *Every size- k Boolean antichain in $\mathcal{S}(V)$ corresponds uniquely to:*

1. *a subspace $X \leq V$ of codimension $d \geq k$;*
2. *an unordered direct-sum decomposition*

$$V/X = U_1 \oplus \cdots \oplus U_k$$

into nonzero subspaces.

Hence, for $k \geq 1$,

$$|\mathcal{C}_k(\mathcal{S}(\mathbb{F}_q^n))| = \sum_{d=k}^n \begin{bmatrix} n \\ d \end{bmatrix}_q D_{d,k}(q). \quad (22)$$

Together with $|\mathcal{C}_0| = 1$, this enumerates all Boolean antichains in every finite subspace lattice.

Proof. Let $C = \{H_1, \dots, H_k\}$ be Boolean, put $X = \bigcap_i H_i$, and let $A_i = \bigcap_{j \neq i} H_j$ be the atoms of its Boolean span. In the quotient V/X , set $U_i = A_i/X$. The Boolean identities give

$$\sum_i U_i = V/X, \quad U_i \cap \sum_{j \neq i} U_j = 0,$$

so the sum is direct and every U_i is nonzero. Conversely, from such a decomposition define

$$H_i/X = \bigoplus_{j \neq i} U_j.$$

The sums of subcollections of the U_i form a Boolean sublattice, whose coatoms are the H_i . The two constructions recover X , the summands, and the coatoms, so they are inverse.

There are $\begin{bmatrix} n \\ d \end{bmatrix}_q$ choices for X . For fixed positive dimensions $d_1, \dots, d_k$, the group $\mathrm{GL}_d(q)$ acts transitively on ordered direct-sum decompositions with stabilizer $\prod_i \mathrm{GL}_{d_i}(q)$. Summing over ordered positive compositions and dividing by $k!$ gives Equation (21), then Equation (22). $\square$

For example, the size distribution for the subspace lattice of $\mathbb{F}_2^3$ is

$$1, 15, 49, 28.$$

At $k = n$, Equation (22) reduces to the projective-basis formula in Equation (24) below.

11 Graphic and classical specializations

Let G be a finite graph, allowing loops and parallel edges, and let $c(G)$ be its number of connected components. Its cycle matroid has rank $|V(G)| - c(G)$. Simplifying the cycle matroid deletes loops and identifies parallel edges.

Corollary 11.1 (Spanning forests). *The maximum Boolean antichains in $\mathcal{L}(M(G))$ are in bijection with the spanning forests of the simplification of G . Their size is $|V(G)| - c(G)$. If G is simple, they are in bijection with the spanning forests of G ; if G is also connected, they are in bijection with its spanning trees.*

With an edge variable x_e , Equation (13) is the spanning-forest polynomial of a multigraph: each simplified edge contributes the sum of the variables in its parallel class.

Proof. The bases of a cycle matroid are the spanning forests having one spanning tree in each connected component. Apply Theorems 6.1 and 7.1. $\square$

Taking $G = K_n$, the flat lattice is the partition lattice and Cayley's formula gives n^{n-2} maximum Boolean antichains, recovering the result of Garber et al. [1].

Every rank-three flat lattice is already completely enumerated by the one-element, pair, and maximum theorems:

$$\begin{aligned} c_0(M) &= 1, & c_1(M) &= |\mathcal{L}(M)| - 1, \\ c_2(M) &= \frac{1}{2} \left(\sum_{X \leq E} \mu(X, E) \ell(X)^2 - 2|\mathcal{L}(M)| + 1 \right), & c_3(M) &= b(\text{si}(M)). \end{aligned}$$

All higher coefficients vanish. In particular, let $P(q)$ be the flat lattice of any finite projective plane of order q , and put $N = q^2 + q + 1$. Counting pairs of lines, nonincident point-line pairs, and noncollinear point triples gives the complete distribution

$$(c_0, c_1, c_2, c_3) = \left(1, 2N + 1, \binom{N}{2} + Nq^2, \binom{N}{3} - N \binom{q+1}{3} \right). \quad (23)$$

This calculation uses only the incidence axioms and therefore includes non-Desarguesian finite projective planes.

For $r \geq 1$ and the projective geometry $PG(r-1, q)$, an unordered basis is an unordered set of r independent projective points. Hence

$$|\mathcal{C}_r(PG(r-1, q))| = \frac{1}{r!} \prod_{i=0}^{r-1} \frac{q^r - q^i}{q - 1}, \quad r \geq 1. \quad (24)$$

12 Scope of the matroid correspondences

The global classification is unconditional for finite lattices. Its matroid correspondences take particularly precise forms.

Parallel classes. The unweighted maximum-antichain correspondence naturally counts bases of the simplification. For a rank-two matroid with parallel classes $\{a, a'\}$ and $\{b, b'\}$, the flat lattice has one maximum Boolean antichain, whereas the original matroid has four bases. The weighted formula recovers the latter count as $4 = 2 \cdot 2$.

Graphic specialization. For a disconnected graph, the relevant size is the matroid rank $|V(G)| - c(G)$ and the corresponding objects are spanning forests. Connected simple graphs specialize this statement to spanning trees, as in Corollary 11.1.

Off-diagonal layers.

Contraction bases characterize the rank-tight diagonal. The global classification also includes smaller non-rank-tight Boolean antichains. In $U_{3,4}$, for example, the two flats

$$\{1, 2\}, \quad \{3, 4\}$$

form a Boolean antichain of size two: their meet is empty and their join is the top. Its bottom has rank zero, whereas rank-tightness would require rank one. Theorem 3.1 detects this span through its atom-reconstruction gaps and, more generally, enumerates every off-diagonal layer.

13 Further directions

Equation (5) determines the complete two-parameter enumerator by antichain size and bottom rank. The rank-tight diagonal reduces to contraction bases, while the off-diagonal terms record realized strict polymatroid profiles. A natural next question is whether substantial parts of this profile distribution factor through familiar matroid invariants, or whether it supplies explicit separations beyond the Tutte polynomial.

The source paper also asks about the partial order on Boolean antichains. Under Theorem 6.1, any natural order on maximum Boolean antichains becomes an order on simplified matroid bases. Determining whether it agrees with a known basis order for special matroid classes opens a further structural direction.

Formalization. The accompanying Lean 4 development derives every formally mapped result from `mathlib` under a kernel-only trust boundary and has passed per-declaration axiom audits.

Code and formalization availability.

The Lean 4 sources, theorem map, verification instructions, and finite regression scripts are available at `github.com/crabsatellite/matroid-boolean-antichains`.

Declaration of Generative AI and AI-assisted technologies in the writing process.

During the preparation of this work, the author used an author-developed local implementation of the Proof Engine framework for AI-assisted mathematical research. The implementation operates through replaceable AI-agent and harness components; the present work used OpenAI Codex within that implementation. The framework is described in *Proof Engine Infrastructure* [2]. The author reviewed and edited the resulting material and assumes full responsibility for the definitions, statements, proofs, citations, code, and exposition.

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